

Lab 8: Compare Distributions with Boxplots

Level: Intermediate Mapping: LO4 | K5 | A3 | A4 | A7 | A10 Concepts: boxplot, quartiles, outliers, group comparison

Scenario

An agriculture team compares chick weights across feed groups.

Goal

A grouped boxplot, descriptive table and visual interpretation.

Files in this folder

- lab-08.R - completed guided R script
- verify.R - repeatable acceptance check
- lab-workbook.xlsx - brief, input data, controls and evidence record
- input-data.csv - portable fallback if readxl is unavailable

Detailed step-by-step

1. Import the feed dataset from the workbook.
2. Confirm feed is a factor and count observations per group.
3. Create a grouped boxplot with readable labels.
4. Calculate median and IQR for every feed.
5. Identify one unusual group and document the evidence.

Run and verify

1. Set the RStudio working directory to this lab folder.
2. Open lab-08.R and read the header, inputs and validation rules.
3. Run the complete script with `source("lab-08.R")`.
4. Confirm the Console ends with LAB VERIFIED: outputs created and checks passed.
5. Run `source("verify.R")` and retain the verification message.
6. Open analysis_output.csv and compare the evidence with the workbook Control sheet.
7. Complete the workbook Evidence sheet and save it without changing the filename.

Acceptance criteria

- [] Boxplot includes every feed group.
- [] Summary table agrees with the plotted medians and spread.
- [] Interpretation distinguishes observation from causal claim.

Troubleshooting

- there is no package called 'readxl': the script automatically uses input-data.csv; install readxl later if Excel import is required.

- cannot open file: confirm RStudio's working directory is this exact lab folder.
- object not found: run the script from the top so imports and validations occur before analysis.
- Verification fails: inspect the first error, compare column names with Input_Data, and rerun from a clean session.

Evidence to submit

Completed Excel Evidence sheet, R script, generated output files and the final Console verification message.